

Complete Local-Dimension Universality at Countable Convolution Fibres

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Status. Candidate full result, likely valid, pending expert review. For every $0 < \lambda \leq 1$ and every prescribed finite $\beta \geq 0$, there are compactly supported probability measures μ, ν on $\mathbb{R}$ such that $\underline{\dim}_{\text{loc}} \mu \geq \lambda$ everywhere, the representation fibre over zero is exactly countably infinite, every fibre point has product lower local dimension at least λ , and $\dim_{\text{loc}}(\mu * \nu)(0) = \beta$. Thus countability imposes no numerical restriction at all. A genuinely weaker scale-averaged replacement for the source paper’s uniform hypothesis is also proved.

1 Source questions and resolution

For a finite Borel measure η on a metric space and $x \in \text{supp } \eta$, write

$$\underline{\dim}_{\text{loc}} \eta(x) = \liminf_{r \downarrow 0} \frac{\log \eta(B(x, r))}{\log r}, \quad \overline{\dim}_{\text{loc}} \eta(x) = \limsup_{r \downarrow 0} \frac{\log \eta(B(x, r))}{\log r}.$$

When they agree, their common value is $\dim_{\text{loc}} \eta(x)$.

Hare and Prandi [1] prove lower-local-dimension inheritance for convolution under a common-scale almost-everywhere hypothesis. In Section 5, PDF page 12, they ask:

“Is there a weaker or different condition for Theorem 2.7. If we know $\underline{\dim}_{\text{loc}} \mu(x) \geq \lambda$, under what conditions is it true that $\underline{\dim}_{\text{loc}}(\mu * \nu)(z) \geq \lambda$.”

They then distinguish finite representation fibres from fibres containing an open family and ask:

“What happens if there are countably many $x \in \text{supp } \mu$ and $y \in \text{supp } \nu$ such that $x + y = z$?”

Theorem 1 gives a complete negative resolution of the countable-fibre question in the lower-local-dimension setting: every finite nonnegative value is possible, even under the pointwise hypothesis and a pairwise product-dimension lower bound. Proposition 1 answers the first question constructively by replacing the common-scale condition with decay of the mass of bad centres; Proposition 2 proves the replacement is strictly weaker.

Is there a weaker or different condition for Theorem 2.7? If we know $\dim_{loc}\mu(x) \geq \lambda$, under what conditions is it true that $\dim_{loc}\mu * \nu(z) \geq \lambda$.

We worked with groups and the real line. Our results from groups translate to $\mathbb{R}^n$ (or vector spaces in general). On the other hand, the results from the real line strongly use the underlying structure of $\mathbb{R}$. This causes some problem when we want to work on $\mathbb{R}^n$. We could define the gaps in $\mathbb{R}^n$ as follows:

Definition 5.1. Let ν be a measure on $\mathbb{R}^n$ of bounded support. We say $A \subset \text{hull}(\text{supp } \nu)$ is a gap in $\text{supp } \nu$ if

- A is connected.
- A is open.
- $A \cap (\text{supp } \nu) = \emptyset$

In contrast to working in $\mathbb{R}$, the gaps in $\mathbb{R}^n$ can have any shape. In addition, to be able to cover any gap, we need the support of μ to be a ball. Then we could recover the results of Section 3 but with the $z \in B(0, 1) + \text{supp } \nu$.

The results of Section 4 would still be highly dependent on the shape of the gaps. When working on $\mathbb{R}^n$, there is the question of what happens to points $z \in \text{supp } \mu * \nu$, such that $z \in \partial \text{supp } \mu + \partial \text{supp } \nu \setminus (\text{int}(\text{supp } \mu) + \text{supp } \nu)$. Theorem 4.1 answers the question of what happens in $\mathbb{R}$, if the supports of both μ and ν are convex. In $\mathbb{R}^n$, even if the supports are convex, things get complicated. As an example, let $\text{supp } \mu = \text{supp } \nu = [0, 1] \times [0, 1]$. Then $\text{supp } (\mu * \nu) = [0, 2] \times [0, 2]$. If $z \in (0, 2) \times \{0\}$, it is not clear what a good bound for $\dim_{loc}(\mu * \nu)(z)$ would be. Theorem 2.1 gives an upper bound; but in practice, it is not a good upper bound, since it requires one to calculate a new measure and then the local dimension of that new measure.

From the work we present, we deduce that the “number of points” behind $z \in \text{supp } \mu * \nu$ is key. If $z \in \text{int}(\text{supp } \mu) + \text{supp } \nu$ is a way to say that there are enough points such that $x + y = z$, since we have an open set of points in $\text{supp } \mu$ from where to choose the points that have the properties we want, as we do in theorems earlier in this paper. In other cases where there are finitely many $x \in \text{supp } \mu$ and $y \in \text{supp } \nu$ such as $x + y = z$, then we can work around and do something, as in Section 4. In other words, in the finite case, there are enough constraints that we have something to work with. It is an open question as to what happens in the middle ground between finiteness and contained in an open set. What happens if there are countably many $x \in \text{supp } \mu$ and $y \in \text{supp } \nu$ such that $x + y = z$?

Figure 1: The two source questions in Section 5 of Hare–Prandi [1], PDF page 12.

2 Proof intuition

Countability does not provide quantitative uniformity. A countable fibre can be assembled from tiny packets at geometrically separated centres. The packet weights determine the mass delivered to the target, while the packet widths determine the scale at which it arrives. These two parameters can be tuned independently without changing the eventual pointwise dimension at any fixed packet.

For target dimensions $0 \leq \beta \leq 1$, one measure consists of symmetric uniform packets and the other of aligned atoms. The n th aligned contribution has mass p_n^2 and width ℓ_n . Choosing $p_n^2 \asymp \ell_n^\beta$ makes the aligned convolution mass scale like r^β . A different super-exponential choice of widths yields dimension zero at every sufficiently small scale, not merely along a subsequence.

For $\beta > 1$, atoms would force a linear contribution, so both factors are made one-sided near each representation pair. Power-law endpoint densities with exponents a and b convolve to order r^{a+b} ; setting $a + b = \beta$ gives the target. Geometric separation forces every off-diagonal interaction into a small tail. Choosing the tail exponent $D > \beta/2$ makes its total mass $O(r^{2D}) = o(r^\beta)$.

The positive criterion uses a complementary idea. At each scale, centres where the desired Frostman estimate fails are allowed, but their ν -mass must decay with the same power as the target ball mass. Splitting the convolution integral into good and bad centres then gives the desired bound.

3 A common separation estimate

The following elementary observation will control every unintended interaction. All constants below may depend on the fixed construction parameters, never on r .

Lemma 1 (Off-diagonal tail bound). *Let $x_{n+1} \leq x_n/4$ and let J_n be an interval contained in $[x_n - x_n/16, x_n + x_n/16]$. Give J_n mass p_n , where the p_n form a geometric sequence. Pair these packets with packets or atoms lying within $x_n/16$ of $-x_n$ and also having mass p_n .*

If an off-diagonal pair of indices $k \neq n$ contributes to a ball $B(0, r)$ under addition, then

$$x_{\min\{k, n\}} \leq 2r.$$

Consequently its total contribution is at most

$$\left(\sum_{x_j \leq 2r} p_j \right)^2.$$

If $p_j = x_j^D$, up to a fixed normalization, this is $O(r^{2D})$.

Proof. Put $j = \min\{k, n\}$, so x_j is the larger centre. The centre separation is at least $3x_j/4$. The relevant packet radii have sum at most $x_j/8$. Thus meeting $B(0, r)$ forces $r \geq x_j/2$. Both indices therefore belong to the displayed tail. Summing all products $p_k p_n$ over that tail gives the square bound. For geometric p_j the tail is comparable to its first term; if $p_j \asymp x_j^D$, it is $O(r^D)$. $\square$

4 Complete universality

Theorem 1 (Countable-fibre universality). *Fix $0 < \lambda \leq 1$ and a finite $\beta \geq 0$. There are compactly supported Borel probability measures μ, ν on $\mathbb{R}$ such that:*

- (i) $\underline{\dim}_{\text{loc}} \mu(x) \geq \lambda$ for every $x \in \text{supp } \mu$;

(ii) *the representation fibre at zero is exactly*

$$\mathcal{F}_0 = \{(x, y) \in \text{supp } \mu \times \text{supp } \nu : x + y = 0\} = \{(0, 0)\} \cup \{(x_n, -x_n) : n \geq 1\};$$

(iii) *every point of $\mathcal{F}_0$ has product-measure lower local dimension at least λ , for any standard product metric;*

(iv) *the local dimension of the convolution exists and satisfies*

$$\dim_{\text{loc}}(\mu * \nu)(0) = \beta.$$

4.1 The range $0 < \beta \leq 1$

Choose $0 < q < 1/16$ and let

$$p_n = (1 - q)q^{n-1}, \quad x_n = p_n \quad (n \geq 1).$$

Thus $\sum p_n = 1$. Choose $\tau > 0$ small and set

$$\ell_n = \tau p_n^{2/\beta}, \quad I_n = [x_n - \ell_n, x_n + \ell_n],$$

with $\ell_n \leq x_n/16$. Define

$$\mu = \sum_{n=1}^{\infty} \frac{p_n}{2\ell_n} \mathbf{1}_{I_n} dx, \quad \nu = \sum_{n=1}^{\infty} p_n \delta_{-x_n}. \quad (1)$$

The intervals are disjoint and accumulate only at zero. The measure μ has local dimension one at every point of every interval, including its endpoints. Since packet locations and their masses are both comparable to p_n , geometric-tail comparison gives

$$\mu(B(0, r)) \asymp r, \quad \nu(B(0, r)) \asymp r.$$

Hence both measures have local dimension one at zero. Their supports show that the fibre is exactly the set in (ii). Its nonzero points have product dimension $1 + 0 = 1$, and $(0, 0)$ has product dimension two. Thus (i)–(iii) hold for every $0 < \lambda \leq 1$.

Write $\eta = \mu * \nu$. The aligned pairs contribute exactly

$$A(r) = \sum_{n=1}^{\infty} p_n^2 \min \left\{ 1, \frac{r}{\ell_n} \right\}. \quad (2)$$

Indeed, translating the n th uniform packet by $-x_n$ centres it at zero. Since

$$p_n^2 = \tau^{-\beta} \ell_n^\beta$$

and the ℓ_n form a geometric sequence, choose $N = N(r)$ so that $\ell_N \leq r < \ell_{N-1}$. The full aligned tail gives

$$A(r) \geq \sum_{n \geq N} p_n^2 \geq cr^\beta.$$

For $0 < \beta < 1$, the partial-packet geometric sum and the full tail give

$$A(r) \leq Cr^\beta.$$

For $\beta = 1$, each of the first $N - 1$ partial packets contributes a fixed multiple of r , and hence

$$A(r) \leq CN(r)r, \quad N(r) = O(|\log r|).$$

By Lemma 1, all off-diagonal pairs contribute $O(r^2)$. It follows that, for $0 < \beta < 1$,

$$cr^\beta \leq \eta(B(0, r)) \leq Cr^\beta,$$

while for $\beta = 1$,

$$cr \leq \eta(B(0, r)) \leq C(1 + |\log r|)r.$$

In both cases, division by $\log r$ proves $\dim_{\text{loc}} \eta(0) = \beta$.

4.2 The value $\beta = 0$

Keep p_n, x_n and the measures in (1), but now choose

$$\ell_n = \tau e^{-n^2},$$

with τ small enough that the packets remain disjoint and $\ell_n \leq x_n/16$. If $\ell_N \leq r < \ell_{N-1}$, the fully captured aligned tail gives

$$\eta(B(0, r)) \geq \sum_{n \geq N} p_n^2 \geq cq^{2N}.$$

At the same time $|\log r| \asymp N^2$. Since every small-scale exponent is nonnegative,

$$0 \leq \frac{\log \eta(B(0, r))}{\log r} \leq \frac{O(N)}{N^2 + O(1)} \longrightarrow 0.$$

Thus the local dimension exists and equals zero. The support and factor local-dimension arguments above are unchanged.

4.3 The range $\beta > 1$

Choose

$$D > \max\{\lambda, \beta/2\}, \quad a = \lambda, \quad b = \beta - \lambda > 0.$$

Take $q > 0$ so small that $q^{1/D} < 1/16$, set

$$p_n = (1 - q)q^{n-1}, \quad x_n = p_n^{1/D}, \quad \ell_n = \tau p_n^{2/\beta},$$

and choose $\tau > 0$ so that $\ell_n \leq x_n/16$ for all n . This is possible because $2/\beta - 1/D > 0$.

The n th component of μ is supported on $[x_n, x_n + \ell_n]$, has total mass p_n , and in the coordinate $s = x - x_n$ has normalized density

$$\frac{as^{a-1}}{\ell_n^a} \mathbf{1}_{(0, \ell_n)}(s).$$

The n th component of ν is supported on $[-x_n, -x_n + \ell_n]$, has total mass p_n , and in the coordinate $t = y + x_n$ has normalized density

$$\frac{bt^{b-1}}{\ell_n^b} \mathbf{1}_{(0, \ell_n)}(t).$$

At the left endpoint x_n , the local dimension of μ is $a = \lambda$; at every other point of its n th packet it is one. Geometric-tail comparison at zero gives

$$\dim_{\text{loc}} \mu(0) = D.$$

Thus (i) holds. The analogous dimension of ν at zero is also D .

For equal indices, a sum has the form $s+t \geq 0$ and equals zero only when $s=t=0$. For unequal indices, the centre gap is larger than the sum of the packet lengths, so zero is not represented. This proves (ii). The product dimension at $(x_n, -x_n)$ is

$$a+b=\beta \geq \lambda,$$

and the product dimension at $(0,0)$ is $2D > \beta$, proving (iii).

For an aligned pair and $0 < r \leq \ell_n$, direct integration gives

$$\begin{aligned} p_n^2 \int_0^r \frac{as^{a-1}}{\ell_n^a} \left(\int_0^{r-s} \frac{bt^{b-1}}{\ell_n^b} dt \right) ds \\ = p_n^2 a B(a, b+1) \left(\frac{r}{\ell_n} \right)^{a+b}. \end{aligned}$$

For all $r > 0$, the same contribution is bounded above by

$$Cp_n^2 \min \left\{ 1, (r/\ell_n)^\beta \right\}.$$

Since $p_n^2 = \tau^{-\beta} \ell_n^\beta$ and the lengths form a geometric sequence, the total aligned contribution satisfies

$$cr^\beta \leq A(r) \leq C(1 + |\log r|)r^\beta.$$

Lemma 1 bounds the off-diagonal contribution by

$$C \left(\sum_{x_n \leq 2r} p_n \right)^2 \leq Cr^{2D} = o(r^\beta).$$

Therefore

$$cr^\beta \leq (\mu * \nu)(B(0, r)) \leq C(1 + |\log r|)r^\beta,$$

and $\dim_{\text{loc}}(\mu * \nu)(0) = \beta$. This completes the proof of Theorem 1.

Corollary 1. *For $0 < \lambda \leq 1$, neither the countability of the representation fibre, the pointwise bound $\underline{\dim}_{\text{loc}} \mu \geq \lambda$, nor even the conjunction of these facts with product-dimension lower bounds on every fibre point imposes any nontrivial numerical restriction on $\dim_{\text{loc}}(\mu * \nu)(z)$.*

5 A weaker averaged condition that restores inheritance

Proposition 1 (Bad-centre decay criterion). *Let μ, ν be compactly supported probability measures on $\mathbb{R}$, fix $z \in \text{supp}(\mu * \nu)$, and let $\alpha > 0$. For $0 < \varepsilon < \alpha$ put*

$$E_\varepsilon(r) = \{y \in \text{supp } \nu : \mu(B(z-y, r)) > r^{\alpha-\varepsilon}\}.$$

Suppose that for every $\varepsilon \in (0, \alpha)$ there are $C_\varepsilon < \infty$ and $r_\varepsilon > 0$ such that

$$\nu(E_\varepsilon(r)) \leq C_\varepsilon r^{\alpha-\varepsilon} \quad (0 < r < r_\varepsilon). \quad (3)$$

*Then $\underline{\dim}_{\text{loc}}(\mu * \nu)(z) \geq \alpha$.*

Proof. Split the convolution integral into good and bad centres. Since both measures are probabilities,

$$\begin{aligned} (\mu * \nu)(B(z, r)) &= \int \mu(B(z - y, r)) d\nu(y) \\ &\leq r^{\alpha-\varepsilon} \nu(\mathbb{R} \setminus E_\varepsilon(r)) + \nu(E_\varepsilon(r)) \\ &\leq (1 + C_\varepsilon) r^{\alpha-\varepsilon}. \end{aligned}$$

Taking logarithms and dividing by the negative number $\log r$ gives $\underline{\dim}_{\text{loc}}(\mu * \nu)(z) \geq \alpha - \varepsilon$. Let $\varepsilon \downarrow 0$. $\square$

The source hypothesis makes $\nu(E_\varepsilon(r)) = 0$ below a common scale, so it implies (3). The converse fails.

Proposition 2. *There are compactly supported probability measures $\mu, \hat{\nu}$ for which Proposition 1 applies at $z = 0$ and $\alpha = 1$, but the source paper's common-scale almost-everywhere condition fails.*

Proof. For $n \geq 2$, set

$$x_n = 2^{-n}, \quad \ell_n = 2^{-n^3}, \quad I_n = [x_n - \ell_n, x_n + \ell_n], \quad m_n = 12 \cdot 2^{-2n},$$

and let

$$\mu = \sum_{n=2}^{\infty} \frac{m_n}{2\ell_n} \mathbf{1}_{I_n} dx, \quad \hat{\nu} = Z^{-1} \sum_{n=2}^{\infty} 2^{-n^4} \delta_{-x_n}.$$

Write $r = 2^{-L}$. If $n^3 \leq \varepsilon L/2$ and L is sufficiently large, then

$$\mu(B(x_n, r)) \leq 12 \cdot 2^{\varepsilon L/2} r \leq r^{1-\varepsilon}.$$

Thus every bad atom has $n > (\varepsilon L/2)^{1/3}$, and its total mass is at most

$$C \sum_{n > (\varepsilon L/2)^{1/3}} 2^{-n^4} \leq C' 2^{-c_\varepsilon L^{4/3}} \leq C' r^{1-\varepsilon}.$$

The averaged criterion applies.

At the packet scale $r = \ell_n$, however,

$$\frac{\log \mu(B(x_n, \ell_n))}{\log \ell_n} = \frac{2n \log 2 - \log 12}{n^3 \log 2} \rightarrow 0.$$

Every proposed common cutoff therefore leaves a positive-mass atom that violates the exponent-1/2 estimate. The source condition fails. $\square$

6 Verification, scope, and novelty

The proof is exact. The accompanying checker evaluates truncated versions of both packet models, confirms the fibre-separation inequalities, and recovers the target logarithmic slopes for representative values $\beta = 0, 1/3, 1, 3/2, 2, 4$. These numerical checks are sanity tests only; the geometric-series and beta-integral estimates above are the proof.

Theorem 1 is full for the source's countable-fibre question in the lower-local-dimension setting: every finite nonnegative value occurs, so no finer answer can follow from the stated numerical data and countability alone. It does not classify upper local dimensions on arbitrary groups or settle the source's separate higher-dimensional boundary question.

The run indexes were searched for arXiv:2502.18337, countable representation fibres, lower local dimension under convolution, and scale-wise bad-centre conditions. Bounded web searches on August 9, 2026 used the exact source sentence, title, arXiv id, and combinations of “local dimension,” “convolution,” “countable fibre,” and “universality.” They found the source and Prandi’s 2026 thesis [2], but no later resolution or matching construction. The thesis’s subsequent “addition of measures” chapter concerns the ordinary sum $\mu + \nu$; its convolution chapter retains the source theorem and does not answer the countable-fibre question. Novelty confidence is positive but remains subject to expert bibliographic review.

Human review should focus on the off-diagonal tail lemma, the every-scale lower bound in the $\beta = 0$ construction, and the endpoint beta-integral for $\beta > 1$. The recommended verdict is *candidate full, likely valid*.

References

- [1] K. G. Hare and J. G. Prandi, *Bounding the Local Dimension of the Convolution of Measures*, arXiv:2502.18337 (2025); accepted in Real Analysis Exchange.
- [2] J. G. Prandi, *Iterated Function Systems and the Local Dimension of Measures*, Ph.D. thesis, University of Waterloo (2026), Chapter 4.